Real-Time AI Prediction Platform — Technical Report

Scope

Two-week individual demonstration project. Customer churn is used as an illustrative binary classification problem.

Architecture

FastAPI serves inference from a versioned Random Forest. Inference events are stored in JSONL. A standardized mean-shift monitor measures distribution changes. Streamlit visualizes model information and drift. MLflow logs training metrics.

Reproducibility

Synthetic training data are seeded. The training split is stratified and reproducible. Tests cover basic inference, validation, and security of retraining endpoint.

Delivery

Docker Compose runs the API and dashboard. GitHub Actions checks tests and builds images; optional SSH workflow supports a configured remote deployment.

Operational limitations

No cloud resources or GitHub repository have been created. Scheduled workflow is a scaffold; operational retraining requires a persistent deployment scheduler, updated labels, performance gates and model rollback. Public production access needs TLS and endpoint access control.

Verification

Run pytest -q, docker compose up --build, inspect /health, /docs, /monitor/drift and the Streamlit dashboard. Send at least 20 shifted requests to exercise drift detection.